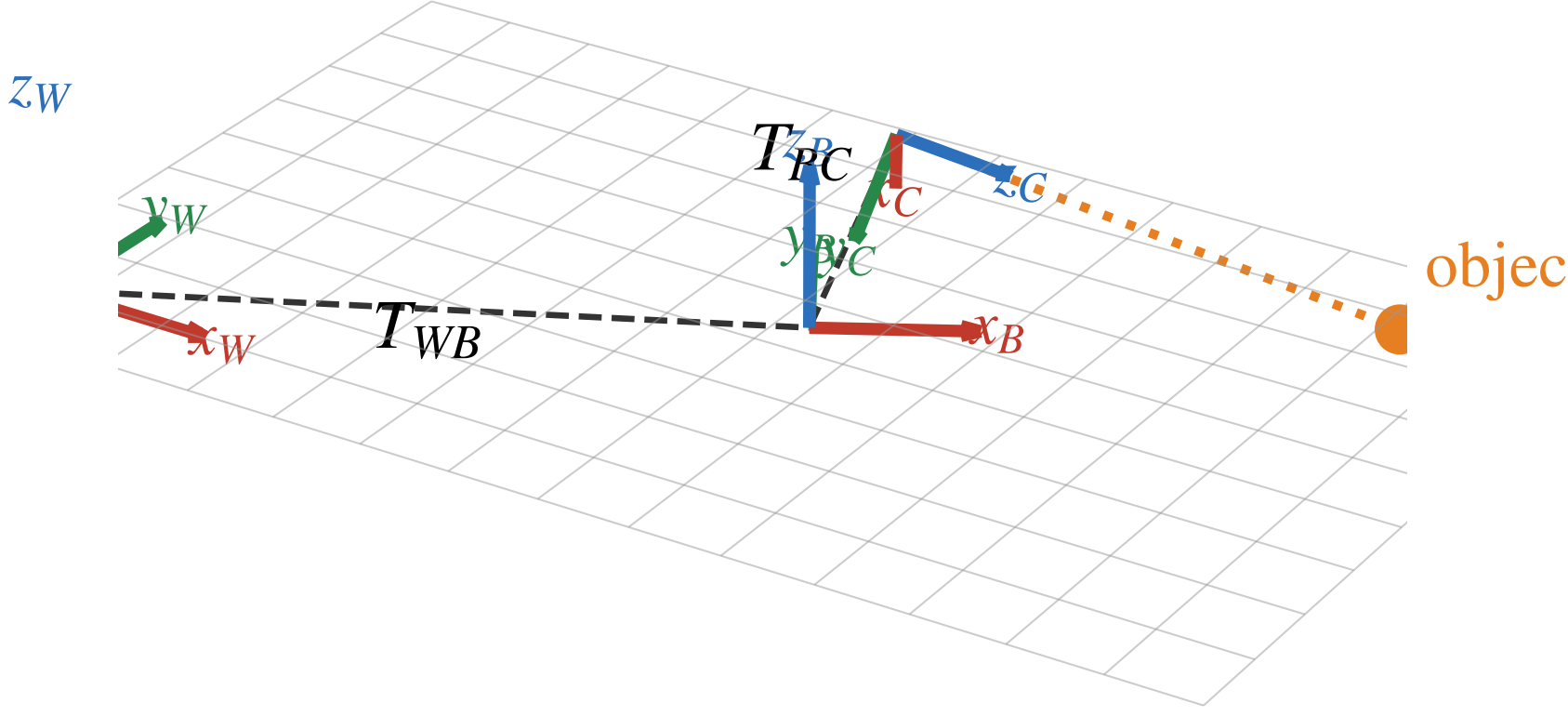World → base → camera → object: ${}^{W}p = T_{WB}\, T_{BC}\, {}^{C}p$